FIN41360 Portfolio and Risk Management

Group Assignment 1

Portfolio Choice and Performance Evaluation

Group 11

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We declare that all material included in this project is the result of our own work and that due acknowledgement has been given in the bibliography and references to all sources beyond, electronic or personal.

Individual contribution

Please outline here below the type of contribution made by each team member (e.g. 45% of the coding, 20% of the writing and gave the presentation) and, in the last column, provide an indicative figure for what percentage of the overall work it represents. This percentage will be treated as a score. Those who will score at or above 25%, 35%, 50% will receive 1, 2 and 4 extra marks (before grade conversion) for their individual contribution, in recognition of their effort.

	Name and surname	**Student nr.**	**Type of contribution**	**Contribution out of 100%**
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2	Eoin Moran	25252950	[INSERT INPUT HERE]	[INSERT INPUT HERE]
3	Jasveen Kaur	25219948	Draft code for Q8 and Q9.	[INSERT INPUT HERE]
4	Kevin O'Regan	25202821	Write-up and coding for Q2, Q4, and Q5; introduction; data and empirical framework section.	20%
5	Tobias Mella	24273004	Writing and coding for Q6 and Q9; first draft of the report up to Q6 in Overleaf; data processing and generation of figures/tables; scheduled and organised meetings.	20%

*Please make sure that the reported figures add up to 100%.

Introduction

Using monthly data from the Kenneth R. French Data Library (French, n.d.), this report studies how estimation risk, asset definition, and implementability change mean-variance portfolio choice. We begin with the 30 industry portfolios over January 1980 to December 2025, compare sample frontiers with Bayes-Stein shrinkage variants, then test whether the main conclusions survive when the opportunity set is changed to individual stocks, augmented with a risk-free asset, or replaced by Fama-French factors and investable ETF proxies. The final part asks whether in-sample winners persist out of sample by comparing Sharpe ratios across two contiguous subperiods and then examines whether simple constraints and shrinkage improve that persistence.

1 Data and Empirical Framework

Following the French data legends, the core inputs are monthly value-weighted 30-industry returns, the one-month Treasury bill rate from the Fama-French factors file, and the FF3/FF5 factor series (French, n.d.). The baseline industry sample runs from 1980-01 to 2025-12. When industries are compared with individual stocks or ETF proxies, each exercise uses the common date overlap so that frontier differences are not driven by mismatched windows. The stock and ETF price histories used for those comparisons were downloaded from Yahoo Finance (Yahoo Finance, n.d.). For the stock exercise, each stock is mapped to an FF30 industry via SIC classification and one representative stock is retained using continuity, availability, and liquidity screens.

We report the global minimum-variance (GMV) portfolio, the tangency portfolio, and the Sharpe ratio. Bayes-Stein shrinkage is used to regularize sample means toward a common target in the spirit of Jorion (1986); where stated, covariance shrinkage is added as a second layer of regularization. Because the factor portfolios are reported in excess-return form, industry and proxy returns are also converted to excess returns when they are compared with FF3 and FF5.

2 Industry Frontier and Opportunity Set Comparisons

2.1 Task 2: 30 Industry Portfolios

Figure 1 plots the monthly efficient frontiers for the 30 industries under three estimators: sample moments, Bayes-Stein mean shrinkage, and Bayes-Stein mean-plus-covariance shrinkage. The ordering is economically intuitive. The sample frontier is the most optimistic because it loads directly on noisy sample means, while the two shrinkage frontiers lie slightly inside it because regularization pulls the optimizer away from extreme return forecasts.

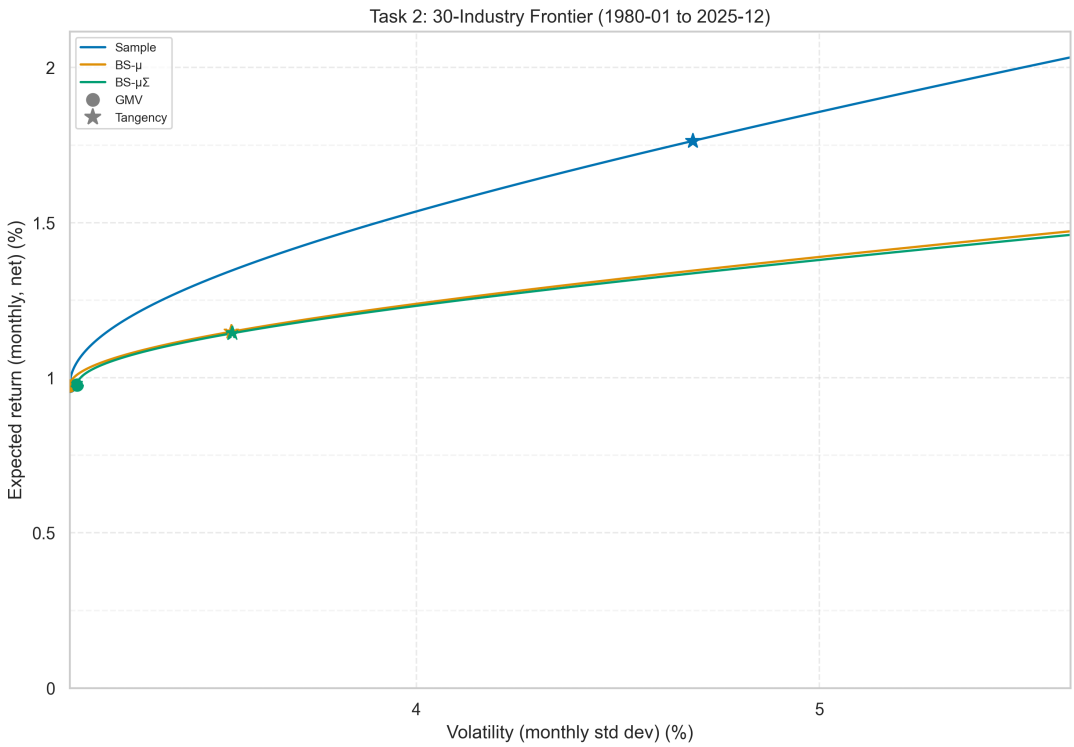


Figure 1: Efficient frontiers for the 30 industry portfolios under sample estimation, Bayes-Stein mean shrinkage, and Bayes-Stein mean-plus-covariance shrinkage. Shrinkage moves the frontier slightly inward, with the largest effect at the tangency portfolio rather than at the GMV portfolio.

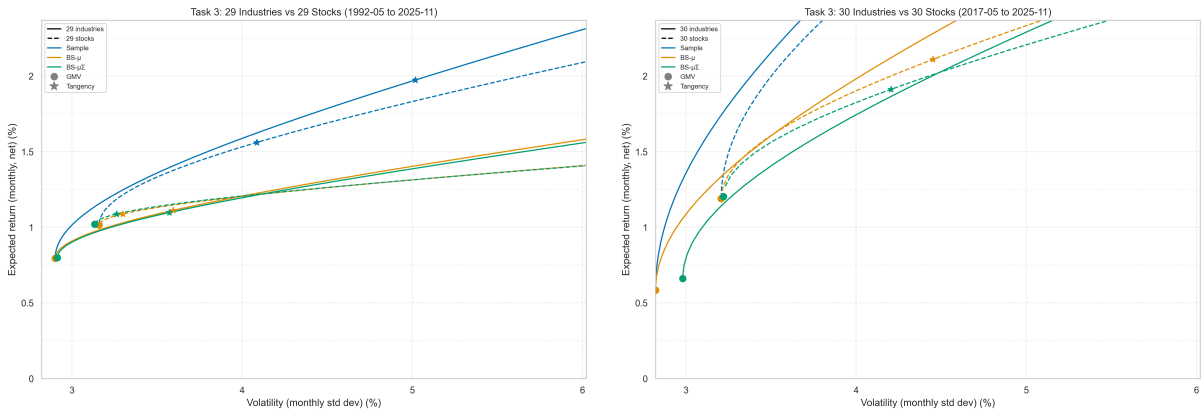
Table 1: Task 2: GMV and tangency portfolio statistics

Portfolio	Estimator	Mean	Excess Return	Volatility	Sharpe
GMV	Sample	0.0097	0.0064	0.0314	0.2050
GMV	BS(μ)	0.0097	0.0064	0.0314	0.2051
GMV	BS(μ, Σ)	0.0098	0.0065	0.0316	0.2059
TAN	Sample	0.0176	0.0143	0.0469	0.3059
TAN	BS(μ)	0.0176	0.0143	0.0467	0.3052
TAN	BS(μ, Σ)	0.0172	0.0139	0.0462	0.3014

Table 1 shows that the GMV portfolio is almost unchanged across estimators, which is expected because it depends mainly on the covariance matrix. Reporting excess returns makes the same point: GMV excess performance is very similar across estimators, while the tangency portfolio remains the higher-Sharpe risky allocation. In this long sample the effect is modest rather than dramatic: Bayes-Stein mean shrinkage leaves tangency Sharpe almost unchanged, and adding covariance shrinkage lowers it only slightly from 0.3059 to 0.3014. The main result is therefore not that shrinkage transforms the opportunity set, but that it mildly reduces in-sample optimism without changing the qualitative ranking of feasible portfolios.

2.2 Task 3: 30 Industries vs 30 Individual Stocks

To make the stock-industry comparison meaningful, both universes are estimated on the same sample window. The main comparison uses the longer common window obtained after dropping Coal, 1992-05 to 2025-11 (403 months); retaining Coal shortens the overlap to 2017-05 to 2025-11 and is treated only as a sensitivity check.



(a) Main comparison on the longer matched window after dropping Coal, 1992-05 to 2025-11

(b) Sensitivity check when Coal is retained, which shortens the common window to 2017-05 to 2025-11

Figure 2: Industry versus stock efficient frontiers on matched sample windows. Both panels use common axis limits, with the x-axis starting at the minimum GMV volatility across the two cases. The longer drop-Coal window is the main comparison, while the with-Coal panel shows how retaining Coal materially shortens the overlap and raises apparent Sharpe ratios for both universes. In the with-Coal case, the sample-estimator tangency portfolios extend beyond the displayed range; see the appendix full-range chart for that view.

In the longer window, industry tangency Sharpe (0.3528) is slightly above the matched stock value (0.3321), while Bayes-Stein and covariance shrinkage leave the gap small. This suggests that the main difference comes from diversification within the pre-formed industry portfolios rather than from estimator choice. The short coal-included sample produces much higher tangency Sharpe for both universes, confirming that overlap design is first-order for interpretation. Because Figure 2 uses common axes for comparability, the with-Coal sample tangency portfolios are partially truncated in that panel and should be read with the appendix full-range chart if the extreme range itself is of interest. The task therefore supports two conclusions: industries remain slightly more efficient than one-stock-per-industry selections on the matched long window, and short windows can overstate the apparent advantage of both universes.

2.3 Task 4: 30 Industries with a Risk-Free Asset

Adding the risk-free asset changes the efficient set from the upper branch of the risky frontier to the capital market line through the tangency portfolio (Lintner, 1965; Sharpe, 1964; Tobin, 1958). In economic terms, investors no longer choose among many risky efficient portfolios; they choose how much to lever or de-lever the same Sharpe-maximizing risky bundle.

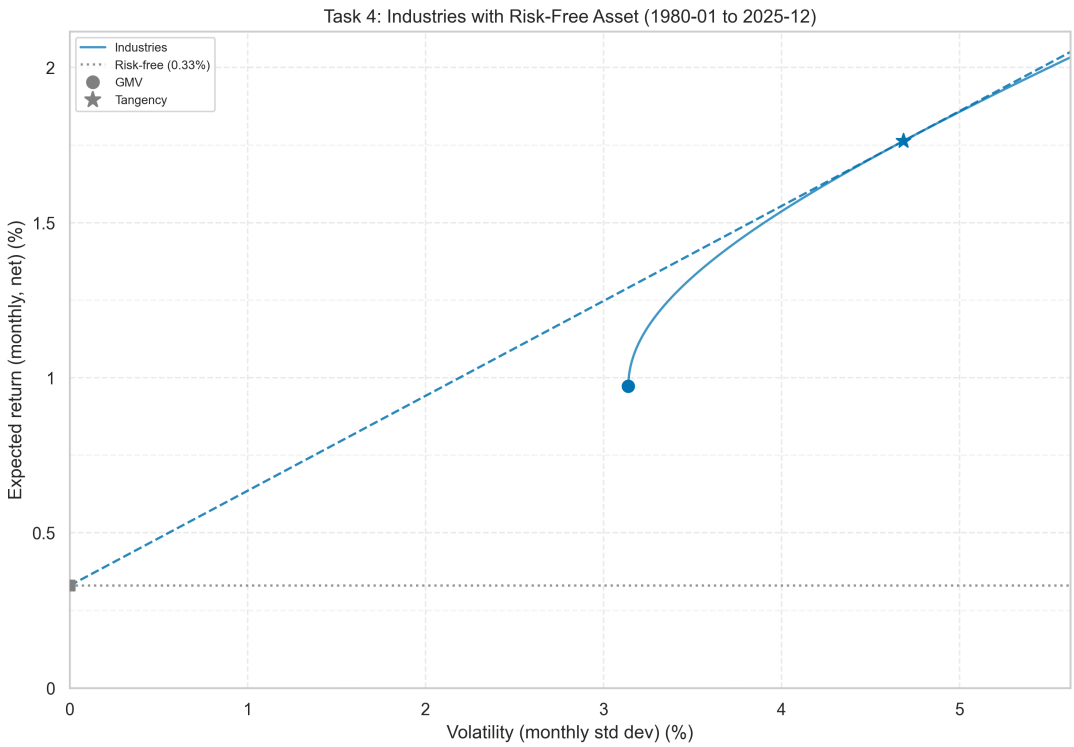


Figure 3: Industry frontier with the risk-free asset added. The capital market line is tangent to the risky frontier at the tangency portfolio, making that portfolio the relevant risky benchmark once borrowing and lending are allowed.

Relative to Task 2, the main implication is that the tangency portfolio becomes the only relevant risky benchmark once borrowing and lending at the average sample risk-free rate are allowed. The risky-only GMV portfolio still exists, but it is no longer the key object for efficient allocation.

3 Factors and Practical Proxies

FF3 and FF5 in excess-return space

To compare industries with Fama-French factors, all series are expressed in excess-return form and estimated over the full 1980-01 to 2025-12 sample (Fama & French, 1993, 2015). Figure 4 shows that FF3 lies well inside the industry opportunity set, whereas FF5 achieves almost the same tangency Sharpe as industries with much lower volatility. This indicates that the five-factor specification spans the relevant return trade-off more efficiently than FF3 in this full sample, even though its expected returns are lower in absolute terms.

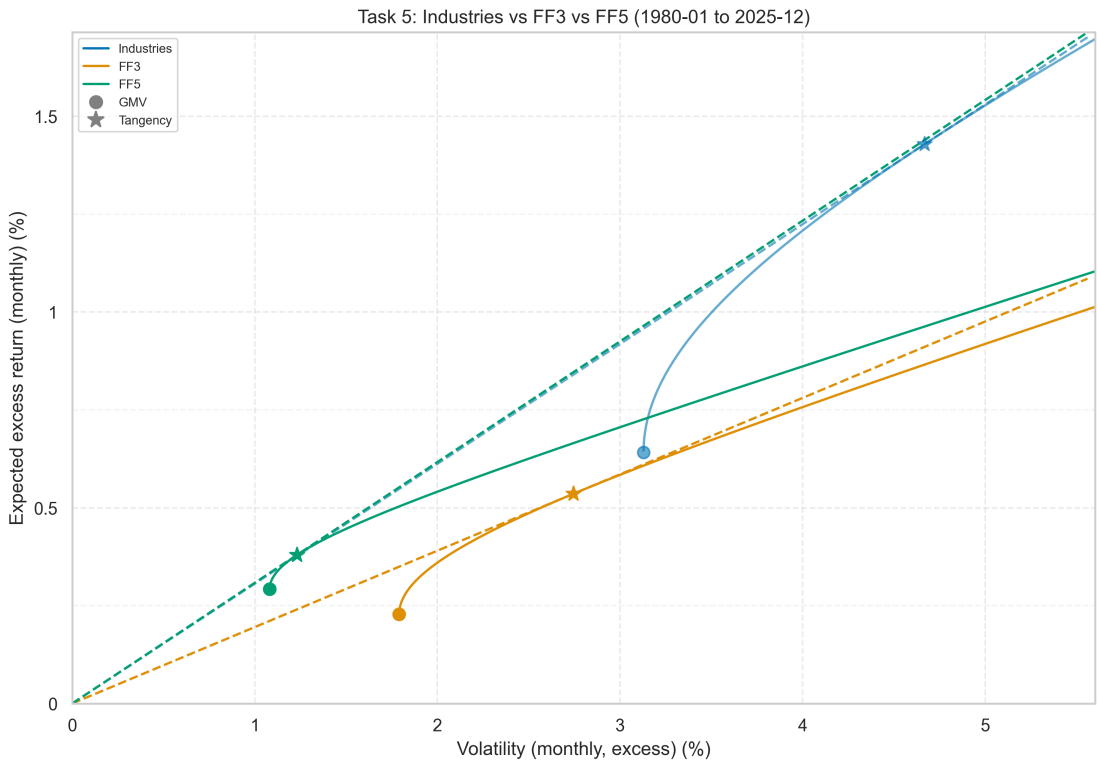


Figure 4: Industry, FF3, and FF5 efficient sets in excess-return space. FF3 lies inside the industry opportunity set, while FF5 delivers a tangency portfolio with a similar Sharpe ratio at much lower volatility.

Table 2: Task 5: tangency comparison in excess-return space, 1980-01 to 2025-12

Universe	Tangency Mean	Tangency Volatility	Tangency Sharpe
Industries	0.0143	0.0467	0.3060
FF3	0.0054	0.0275	0.1951
FF5	0.0038	0.0123	0.3083

Practical proxies

Task 6 is estimated on the shorter common proxy window, 2013-08 to 2025-12, so its levels should not be read as directly comparable to the full-sample Task 5 numbers. Figure 5 and Table 3 therefore compare only factor/proxy pairs within that shorter sample. Proxy5 remains close to FF5 on tangency Sharpe, but Proxy3 diverges more from FF3. The interpretation should be made within the common proxy window: the proxy basket is investable, while the academic factor portfolios remain unconstrained long-short constructs. In particular, the SMB proxy uses IJR as a long-only small-cap implementation, so it does not replicate the academic SMB construction exactly.

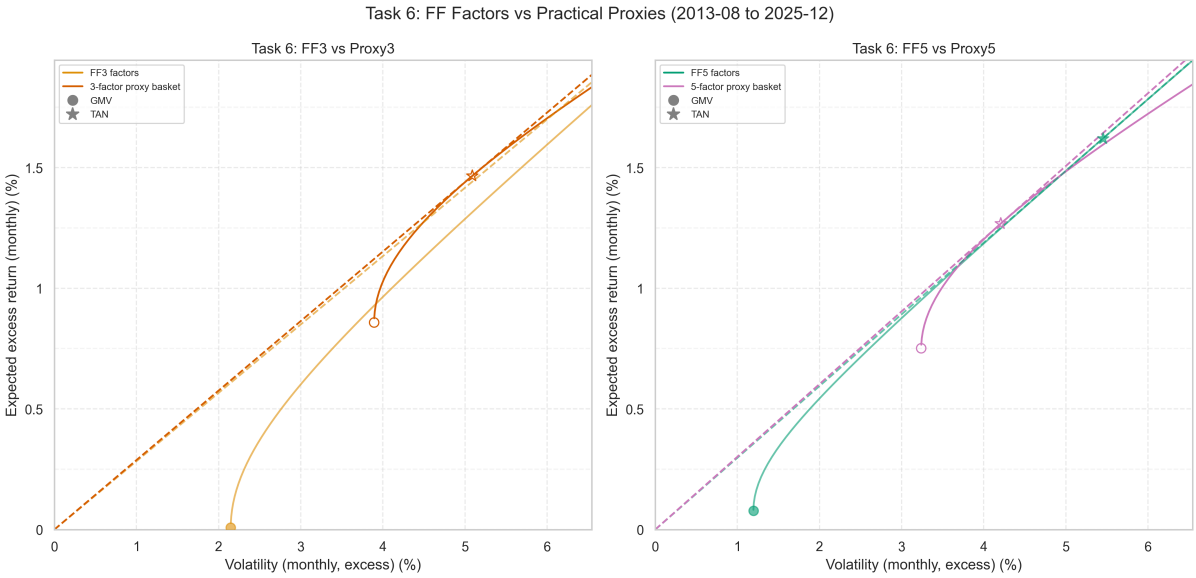


Figure 5: Academic factor frontiers versus practical proxy frontiers in the common proxy window. Proxy5 remains close to FF5, while Proxy3 differs more visibly from FF3 despite achieving a similar tangency Sharpe ratio.

Table 3: Task 6: tangency comparison in the common proxy window, 2013-08 to 2025-12

Pair	Factor Sharpe	Proxy Sharpe
FF3 vs Proxy3	0.2832	0.2880
FF5 vs Proxy5	0.2972	0.3013

The practical implication is narrower than before: in the short proxy window, both proxy baskets preserve tangency Sharpe reasonably well, but Proxy3 reaches that result through a different, more implementable construction. Task 6 is therefore best read as an implementability comparison on a shorter sample, not as a direct continuation of the full-sample Task 5 ranking.

4 Persistence and Constraints

4.1 Task 7: In-Sample vs Out-of-Sample Persistence

The persistence test splits the sample into two contiguous periods, 1980-01 to 2002-12 and 2003-01 to 2025-12, and evaluates the GMV and tangency portfolios chosen in the first period. Table 4 reports in-sample and out-of-sample Sharpe ratios together with Jobson-Korkie and Ledoit-Wolf tests for equality of Sharpe ratios (Jobson & Korkie, 1981; Ledoit & Wolf, 2008). Ledoit-Wolf is treated as the primary decision rule because it is more robust; Jobson-Korkie is reported as a classical cross-check.

Table 4: Task 7: persistence of selected in-sample portfolios

Universe	IS Sharpe	OOS Sharpe	Δ Sharpe	JK p-value	LW p-value	Decision
30 ind TAN	0.3373	0.0813	-0.2560	0.0031	0.0000	Significant IS->OOS deterioration (alpha=0.05)
30 ind GMV	0.1744	0.1650	-0.0093	0.9234	0.9140	No significant deterioration (alpha=0.05)
FF5 TAN	0.3995	0.1987	-0.2008	0.0213	0.0360	Significant IS->OOS deterioration (alpha=0.05)
FF5 GMV	0.3789	0.1782	-0.2007	0.0210	0.0520	No significant deterioration (alpha=0.05)

The answer to the brief is clear. The in-sample tangency portfolios do not remain efficient out of sample in any strong sense: both 30-industry TAN and FF5 TAN suffer statistically significant Sharpe deterioration under Ledoit-Wolf. By contrast, 30-industry GMV does not show significant deterioration, and FF5 GMV is borderline, with Jobson-Korkie rejecting but Ledoit-Wolf not rejecting at the 5% level. That disagreement is not especially surprising: Jobson-Korkie is more dependent on iid normal-return approximations, whereas Ledoit-Wolf is designed to be more robust when those assumptions are doubtful (Jobson & Korkie, 1981; Ledoit & Wolf, 2008). Persistence is therefore much stronger for conservative low-risk portfolios than for high-Sharpe in-sample tangency picks.

4.2 Task 8: Can Constraints and Shrinkage Improve Persistence?

From an investment-management perspective, Task 8 matters because unconstrained mean-variance portfolios can be statistically fragile and economically non-robust even when they look attractive in sample. We therefore test whether a simple no-short-selling constraint, and then Bayes-Stein shrinkage as a cross-check, produce more credible out-of-sample behavior (Jagannathan & Ma, 2003).

The full-sample constraint results show why this extension is useful. For FF3, unconstrained tangency Sharpe is 0.2832, but it relies on gross leverage of 74.66; imposing $w_i \geq 0$ reduces Sharpe to 0.2387, but it also removes the extreme leverage that signals sensitivity to estimation error. The issue is therefore not only implementability: the unconstrained FF3 solution is also less likely to retain its in-sample Sharpe out of sample. The same constraint leaves FF5 with a relatively small Sharpe loss, from 0.2972 to 0.2726, which already suggests that FF5 is structurally more stable than FF3 or Proxy5.

Table 5: Task 8: constrained persistence under $w_i \geq 0$

Universe	IS Sharpe	OOS Sharpe	Δ Sharpe	LW p-value	Decision
Industries	0.2219	0.1995	-0.0224	0.7990	No significant deterioration (alpha=0.05)
FF5	0.3059	0.2619	-0.0440	0.7480	No significant deterioration (alpha=0.05)
Proxy5	0.3969	0.1324	-0.2645	0.0250	Significant IS->OOS deterioration (alpha=0.05)

Table 5 confirms that the no-short constraint improves persistence selectively. In the constrained run, Industries and FF5 show no significant in-sample to out-of-sample Sharpe deterioration, whereas Proxy5 still deteriorates sharply. More broadly, the no-short constraint helps not only because it is easier to implement, but because it regularizes the optimization problem by ruling out the extreme leverage that often reflects overfitting to noisy expected returns.

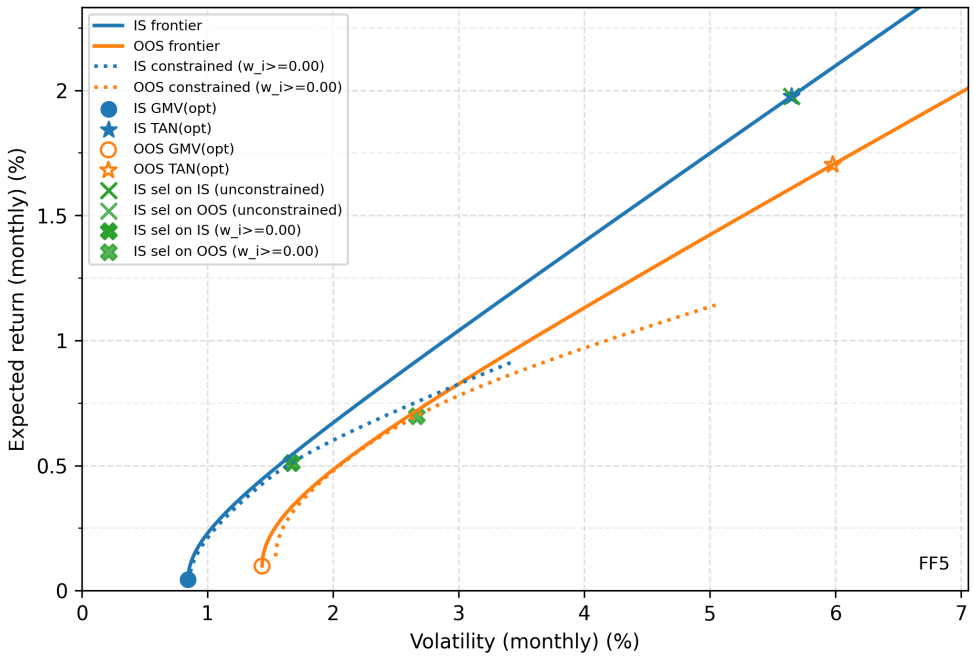


Figure 6: FF5 in-sample and out-of-sample efficient frontiers under the no-short constraint. The selected in-sample portfolios remain close in risk-return space to the out-of-sample efficient set.

Figure 6 adds a more economic interpretation for FF5: within the no-short constrained case, the out-of-sample efficient allocation remains close in risk-return space to a convex combination of the in-sample constrained GMV and tangency portfolios, suggesting that this frontier is comparatively stable across subsamples.

A separate shrinkage sensitivity check points in the same direction but is weaker: for unconstrained tangency portfolios, the industry Sharpe drop improves only modestly from -0.2560 under the sample estimator to -0.2344 under joint mean-and-covariance shrinkage, while FF5 changes very little across shrinkage variants. The extension therefore supports a narrow conclusion: simple constraints do more to improve persistence than shrinkage alone, and the benefit is strongest in the universes that were already less dependent on extreme leverage.

Conclusion

Three findings summarize the report. First, shrinkage moderates in-sample optimism, but in a long sample its effect on frontier shape is modest rather than transformative. Second, sample alignment matters materially: the industry-versus-stock result changes far more with the matched window than with the estimator. Third, constrained portfolios are more credible because they are less sensitive to estimation error and persist better out of sample, with the strongest evidence coming from industries and FF5.

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